SWE 363 · ASSIGNMENT 4

Personal Web Application

A polished portfolio web app showcasing
full-stack, AI, data, and client-based project work.

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SWE 363 — Web Engineering & Development

OBJECTIVE

What I Built

- A polished, responsive personal portfolio web application
- Built with vanilla HTML, CSS, and JavaScript only — no frameworks
- Showcases project-based and client-based experience
- Ready for deployment on GitHub Pages
- Includes full documentation and presentation materials

> **One sentence:** a real portfolio I can use after this course ends.

MOTIVATION

Why This Project Matters to Me

- Bring all my project work into one professional place
- Show real client-based experience without exposing private clients
- Practice production-quality engineering, not just coursework
- Build something I can share with future employers and clients

OVERVIEW

Sections at a Glance

Hero
Strong professional positioning & call to action.

About
Personal summary and six skill groups.

Experience
Project & client work across 7 areas.

Projects
Search, filter, sort, and favorite — dynamic.

GitHub API
Live repositories with retry on failure.

Assistant
Rule-based agentic AI–inspired feature.

Contact
Validated form with draft auto-save.

Documentation
Cards covering all repo documents.

Presentation
What this video covers, summarized.

EXPERIENCE

Real Project-Based Work

🎓 Educational Platforms

Complete platform-style solutions for educational company needs.

💼 Freelance Full-Stack

Responsive websites delivered to different clients.

📊 Client Dashboards

Dashboard-style layouts with cards, filters, and structure.

📈 US Market Tools

Software & data exploration — **not financial advice**.

🤖 Agentic AI Solutions

AI-assisted workflows for automation and decision-making.

🗄️ Database-Driven Systems

Structured data models & CRUD-style logic.

PROJECT SPOTLIGHT

Educational Platform Development

- Built complete platform-style solutions for educational company needs
- Clean structure, organized content, responsive design
- Focused on usability for both learners and administrators
- Iterated based on feedback from real users
- Delivered practical solutions for educational workflows

Platform Architecture | Responsive Design | Dashboard UI | Content Organization

PROJECT SPOTLIGHT

US Market Investment Tools

- Custom stock filtering and scanning interfaces
- Dashboard-style views for market data
- Comparison and decision-support layouts
- Future direction: real market data APIs and visual summaries

> **Important:** these are **software and data exploration tools** — not financial advice or investment recommendations.

PROJECT SPOTLIGHT

Agentic AI Workflow Solutions

- AI-assisted automation for client workflows
- Agent-style guided interactions
- Decision-making support through structured logic
- Responsible, explainable AI usage
- Connects AI concepts with practical software

> Tied to the in-website **Portfolio Assistant** — an agentic AI–inspired rule-based feature.

ARCHITECTURE

Built with Vanilla Web Tech

- **HTML5** — semantic structure
- **CSS3** — design tokens, light/dark theme, responsive grid
- **Vanilla JavaScript (ES6+)** — modular, no frameworks
- **GitHub REST API** — live repository data
- **localStorage** — persistent state

3 Files. Zero Dependencies.

```
index.html
├── css/styles.css
└── js/script.js
```

FEATURE

Dynamic Projects System

- Source of truth: a single **PROJECTS** array in JavaScript
- Search by keyword across title, category, tech, and highlights
- Filter by category
- Sort newest / oldest / A–Z / Z–A
- Favorite a project — persisted in **localStorage**
- Empty state appears when no projects match

FEATURE

Live GitHub API Integration

- Fetches public repos from the GitHub REST API
- Loading skeletons → success → error → retry
- Handles missing descriptions and rate limits gracefully
- ARIA-live status updates for accessibility
- Retry button if the API fails

> **Goal:** the website never breaks, even when the network does.

FEATURE

Contact Form & localStorage

- Client-side validation with inline error messages
- Live character counter on the message field
- Drafts auto-save to localStorage as you type
- Drafts restore automatically after page reload
- Graceful handling if storage is disabled

RESPONSIBLE AI

How I Used AI

- Used ChatGPT / Claude for planning, drafting, and debugging ideas
- Reviewed, edited, and tested every suggestion
- No unmodified AI output was submitted
- Avoided exposing private client details
- Avoided financial advice language for the US market tools

> Full prompt-by-prompt log available in **docs/ai-usage-report.md**.

CHALLENGES

Hardest Parts & How I Solved Them

Designing Without a Framework

Built a small reusable design system using CSS variables.

Handling API Failures

Added explicit loading, error, and retry states.

Describing Client Work Honestly

Used general phrasing instead of inventing names.

Keeping It Accessible

ARIA-live regions and reduced-motion support.

FUTURE WORK

What's Next

- Real backend so the contact form actually sends mail
- Real US market data API for the InvestVision project
- Per-project case-study pages with screenshots
- Privacy-respecting analytics
- Arabic / English internationalization

THANK YOU

One polished portfolio, ready to grow with me.

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Built with HTML, CSS, JavaScript, and responsible AI assistance.